

SAI PRANEETH KAMISHETTY

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PROFESSIONAL SUMMARY

- Applied Generative AI Engineer with 5+ years of experience designing and building LLM-powered applications, retrieval-augmented generation (RAG) systems, AI agents, conversational AI platforms, and intelligent automation workflows across healthcare, analytics, and enterprise environments. Experienced in architecting end-to-end GenAI systems involving semantic retrieval pipelines, embeddings-based search, prompt engineering, contextual memory workflows, structured outputs, tool-calling architectures, and AI-assisted orchestration pipelines using Python, FastAPI, OpenAI APIs, Claude, LangChain/LangGraph workflows, PostgreSQL, Supabase, and modern AI deployment ecosystems.
- Strong background in AI product engineering involving conversational interfaces, AI-driven user experiences, scalable backend orchestration systems, retrieval optimization workflows, metadata-aware retrieval, and rapid AI product prototyping using Vercel, GitHub Actions, n8n, and cloud-native deployment pipelines. Skilled in machine learning, transformer architectures, NLP, backend engineering, and workflow automation with a focus on building production-oriented AI systems solving operational, conversational, and productivity-focused use cases.

CORE SKILLS

- **Generative AI & LLM Engineering:** LLMs, Generative AI, Retrieval-Augmented Generation (RAG), AI Agents, Agentic AI, Multi-Agent Workflows, Prompt Engineering, Tool Calling, Structured Outputs, Conversational AI, Context Management, Prompt Chaining, Prompt Templates, Conversational Memory Handling, Token Optimization, AI Workflow Orchestration
- **RAG, Retrieval & NLP:** Embeddings, Semantic Search, Vector Search, Retrieval Pipelines, Metadata-Aware Retrieval, Hybrid Retrieval Concepts, Context Compression, Chunking Strategies, Query Optimization, Information Retrieval, Contextual Prompt Augmentation, NLP, Text Classification, Semantic Similarity Search
- **Programming & Backend Engineering:** Python, SQL, R, FastAPI, REST APIs, Async API Workflows, JSON Pipelines, API Integration, Backend Workflow Development, Git/GitHub, VS Code, Structured Response Systems
- **Machine Learning & Deep Learning:** Machine Learning, Deep Learning Fundamentals, Transformer Architecture, Regression, Classification, Clustering, Forecasting Models, Statistical Modeling, Neural Networks, Feature Engineering, Model Evaluation, Exploratory Data Analysis
- **AI Workflow Automation:** n8n, Workflow Automation, Intelligent Automation, AI-Based Process Optimization, Conversational Workflow Automation, AI Integrations, Automated Reporting Pipelines
- **Databases & Data Platforms:** PostgreSQL, Snowflake, Supabase, Relational Databases, ETL Pipelines, Data Engineering, Data Transformation, Real-Time Databases, Structured & Unstructured Data Processing
- **Cloud, Deployment & Development Workflow:** Vercel, Supabase, GitHub Actions, Deployment Pipelines, CI/CD Concepts, Cloud Deployment, Full-Stack AI Applications, Rapid AI Prototyping, Version Control Workflows
- **Visualization & Analytics:** Power BI, Tableau, Excel (Advanced), Dashboard Development, KPI Monitoring, Executive Reporting, Data Storytelling, Analytical Reporting
- **Collaboration & Product Engineering:** AI Product Design, Rapid Prototyping, System Design, Cross-Functional Collaboration, Stakeholder Communication, Agile, SDLC, Technical Documentation

EXPERIENCE

Applied Generative AI Engineer, LimeIQ

Dec 2024 Present

- Designed and built LLM-powered workflows and AI-assisted operational systems using OpenAI APIs, conversational AI pipelines, retrieval-augmented generation (RAG), and contextual reasoning workflows supporting healthcare and operational intelligence use cases.
- Architected semantic retrieval pipelines integrating structured and unstructured healthcare datasets using embeddings-based retrieval, chunking strategies, metadata-aware retrieval logic, contextual prompt augmentation, and retrieval-assisted response generation workflows.
- Built conversational AI pipelines involving prompt templates, structured JSON outputs, contextual memory handling, prompt chaining, fallback response systems, and AI-assisted workflow orchestration improving contextual response quality and conversational consistency.
- Developed agentic AI workflows integrating retrieval-assisted reasoning pipelines, planner-style execution flows, contextual memory systems, AI-assisted operational task automation, and tool-calling workflows supporting intelligent operational orchestration.
- Built Python and FastAPI-based backend architectures supporting AI orchestration pipelines, retrieval workflows, contextual search systems, automated reporting architectures, and scalable operational analytics systems.
- Applied transformer-based NLP workflows and semantic retrieval systems to healthcare-related textual datasets supporting contextual information extraction, automated insight generation, semantic similarity search, and AI-assisted search workflows.

- Developed predictive analytics and machine learning workflows involving regression, classification, clustering, forecasting models, and statistical analysis supporting operational optimization and strategic healthcare analytics initiatives.
- Built scalable ETL and data engineering pipelines for ingesting, validating, transforming, and processing heterogeneous healthcare datasets supporting downstream AI workflows and contextual retrieval systems.
- Improved contextual response grounding and retrieval consistency through prompt refinement, context compression workflows, metadata-aware retrieval optimization, and structured response handling systems.
- Developed Power BI and Tableau dashboards visualizing workflow metrics, KPIs, operational insights, retrieval trends, and analytical outputs enabling leadership teams to monitor operational efficiency and strategic performance indicators.
- Conducted workflow optimization and root-cause analysis for operational bottlenecks, reporting inconsistencies, data pipeline inefficiencies, and workflow failures improving automation scalability and operational reliability.
- Collaborated with engineering, analytics, and business stakeholders translating operational requirements into scalable GenAI systems, retrieval workflows, conversational AI systems, and intelligent automation pipelines.
- Created reusable Python automation utilities, backend orchestration workflows, and reporting systems improving operational productivity, AI workflow scalability, and backend processing efficiency.
- Maintained technical documentation for retrieval architectures, AI workflows, prompt systems, orchestration pipelines, feature engineering workflows, and backend analytical systems supporting reproducibility and governance standards.

AI Automation Developer, Info Edge

Dec 2021 Aug 2023

- Built AI-assisted operational systems and contextual analytics workflows integrating NLP pipelines, intelligent reporting architectures, retrieval-based search workflows, and automated backend orchestration systems.
- Developed AI-powered analytical workflows using machine learning, contextual search systems, statistical analysis, and automation pipelines supporting operational intelligence and enterprise decision-making systems.
- Designed preprocessing and semantic retrieval workflows for structured and semi-structured enterprise datasets supporting AI-assisted reporting systems, contextual information retrieval, and operational analytics applications.
- Worked on generative AI experimentation involving conversational reporting concepts, contextual insight generation, retrieval-assisted workflows, and intelligent operational automation systems.
- Built scalable SQL-based ETL pipelines for extracting, transforming, validating, and processing enterprise datasets supporting operational reporting, AI-assisted workflows, and business intelligence systems.
- Applied regression, classification, clustering, forecasting models, exploratory data analysis, and statistical modeling techniques identifying operational trends, workflow inefficiencies, and optimization opportunities across enterprise systems.
- Implemented backend workflow automation systems reducing manual operational processing effort by 65% while improving workflow scalability, reporting consistency, operational efficiency, and analytical reliability.
- Developed Tableau and Excel dashboards visualizing workflow metrics, operational KPIs, reporting trends, and strategic analytical insights supporting enterprise monitoring and operational optimization initiatives.
- Applied preprocessing pipelines, transformation workflows, validation systems, missing-value handling, and data quality monitoring improving downstream analytics performance and reporting reliability.
- Improved reporting scalability and backend processing efficiency through workflow orchestration pipelines, automated validation systems, SQL optimization workflows, and scalable ETL architectures.
- Collaborated with engineering and analytics teams improving AI-assisted reporting architectures, backend orchestration workflows, automation systems, and operational intelligence processes.
- Conducted root-cause analysis for workflow bottlenecks, operational inconsistencies, reporting inefficiencies, and ETL failures implementing corrective actions improving reporting accuracy and operational performance.
- Maintained technical documentation for automation architectures, ETL systems, analytical pipelines, operational workflows, and AI-assisted reporting systems supporting governance and knowledge-sharing initiatives.

Machine Learning & Data Analyst, Wipro

May 2020 Dec 2021

- Built machine learning and AI-assisted analytical workflows supporting enterprise reporting systems, operational optimization architectures, and scalable business intelligence initiatives across multiple business domains.
- Worked on predictive analytics and ML workflows involving anomaly detection systems, forecasting pipelines, classification models, statistical analysis systems, and scalable backend data processing architectures.
- Developed SQL-based data models and Python workflows supporting analytics pipelines, automated reporting architectures, operational monitoring systems, and machine learning experimentation workflows.
- Applied statistical modeling, forecasting techniques, exploratory data analysis, regression workflows, and ML methodologies identifying operational trends and improving enterprise process efficiency by 20%.
- Automated ETL pipelines integrating enterprise datasets into centralized analytical systems improving operational scalability, reporting reliability, workflow efficiency, and data accessibility.

- Developed Tableau dashboards visualizing workflow trends, operational KPIs, analytical metrics, and strategic insights enabling stakeholders to monitor enterprise performance and optimization opportunities.
- Collaborated with cross-functional teams solving operational and analytical problems using AI-assisted workflows, automation systems, statistical models, and data-driven methodologies.
- Prepared analytical reports summarizing forecasting outputs, operational insights, workflow trends, and optimization recommendations supporting stakeholder and leadership decision-making systems.

PROJECTS

AI Voice Receptionist Agent for Restaurant Operations | Python, OpenAI APIs, FastAPI, n8n, Supabase, Vercel

- Built an AI-powered restaurant receptionist using LLM-driven conversational pipelines capable of handling reservations, customer FAQs, business-hour inquiries, contextual conversations, and operational coordination workflows.
- Designed conversational AI architectures involving prompt templates, contextual memory handling, structured outputs, fallback response workflows, contextual retrieval logic, and dynamic response generation improving conversational consistency and interaction quality.
- Built backend orchestration systems using Python, FastAPI, n8n automation workflows, and Supabase real-time databases supporting conversational state handling, backend task routing, and operational workflow automation.
- Integrated cloud-based frontend and backend deployment pipelines using Vercel, GitHub-based deployment workflows, and scalable orchestration architectures supporting rapid AI product iteration.
- Optimized conversational workflows through prompt refinement, contextual retrieval optimization, structured response handling, and context-aware orchestration systems improving response grounding and workflow efficiency.

Agentic RAG Knowledge Assistant & Multi-Agent AI Workflows | Python, LangChain, RAG, Semantic Search, AI Agents

- Built retrieval-augmented generation (RAG) systems using embeddings-based semantic retrieval, document ingestion workflows, chunking strategies, metadata-aware retrieval pipelines, contextual prompt augmentation, and intelligent retrieval workflows.
- Designed multi-agent AI workflows integrating retrieval pipelines, prompt chaining, structured outputs, contextual memory handling, retrieval-assisted reasoning systems, and AI-assisted orchestration architectures.
- Worked on retrieval optimization workflows involving token optimization, metadata-aware retrieval, contextual reranking concepts, prompt refinement, and context compression improving response relevance and grounding quality.
- Built scalable AI workflow prototypes supporting contextual question answering, semantic retrieval, intelligent document understanding, contextual search systems, and AI-assisted operational workflows.
- Developed reusable retrieval architectures and modular orchestration workflows supporting scalable prompt systems and contextual AI-assisted automation pipelines.

Mini Transformer & Deep Learning Research Experiments | Python, Hugging Face Transformers, Deep Learning

- Built and studied lightweight transformer architectures and deep learning workflows to understand attention mechanisms, embeddings systems, tokenization pipelines, contextual reasoning workflows, and foundational large language model architectures.
- Explored prompt engineering strategies, token optimization systems, contextual retrieval workflows, semantic embeddings systems, and efficient AI interaction methodologies improving understanding of modern LLM behavior.
- Conducted experimentation involving transformer pipelines, NLP workflows, contextual memory systems, semantic retrieval concepts, and AI reasoning architectures supporting practical understanding of generative AI systems.
- Developed lightweight experimentation pipelines analyzing transformer inference behavior, contextual reasoning systems, prompt-response patterns, and efficient AI-assisted interaction workflows.

AI TOOLS & DEVELOPMENT ECOSYSTEM

OpenAI APIs, Claude APIs, LangChain, LangGraph, Hugging Face Transformers, Codex, Cursor, Loveable, GitHub Copilot, VS Code, Supabase, Vercel, FastAPI, n8n, Git/GitHub, conversational AI systems, AI-assisted development workflows, prompt engineering, rapid AI prototyping, AI product iteration

EDUCATION

Florida State University

Aug 2023 May 2025

Master of Science, Data Science (Computer Science & Artificial Intelligence Concepts) 3.89

VNR Vignana Jyothi Institute of Engineering and Technology

Aug 2019 May 2023

Bachelor of Technology, EEE 3.68